DIGITAL ELECTRONICS AND COMPUTER ORGANIZATION

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS28	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand different number systems and Boolean Algebra.
2. Design of combinational and sequential logic circuits
3. Understand different Computer Instructions and addressing modes
4. Understand concepts of register transfer logic and arithmetic operations.
5. Learn different types of memory hierarchy

COURSE OUTCOMES

1. Apply minimization techniques to simplify Boolean expressions.
2. Apply the principles of Digital electronics to design combinational and sequential logic circuits.
3. Understand the basics of instruction set and their impact on processor design
4. Illustrate register transfer operations
5. Analyze memory hierarchy and I/O Communication.

UNIT - I	NUMBER SYSTEMS AND BOOLEAN ALGEBRA	CLASSES: 14

Representation of numbers with different radix, Number base conversions, r-1's and r's Complements, Introduction to Binary codes.
Basic Logic Gates, Universal Gates, Basic Theorems and Properties of Boolean algebra, Canonical and Standard forms, K-Map Method (upto 5-Variables)

UNIT - II	COMBINATIONAL AND SEQUENTIAL LOGIC CIRCUITS	CLASSES: 12

Design of Half adder, full adder, half subtractor, full subtractor. Decoder, Encoder, Multiplexer, De-multiplexer and comparator.
Latches and Flip-flops, truth tables, Characteristic table and excitation tables, Conversion of flip-flops, Classification of sequential circuits (synchronous and asynchronous).

UNIT - III	DESIGN OF SEQUENTIAL CIRCUITS AND BASIC COMPUTER ORGANIZATION	CLASSES: 12

Design of Ripple counters, design of synchronous counters, Johnson counter, ring counter, shift registers, bi-directional shift register, universal shift register.
Instruction codes, Computer Registers, Computer Instructions and Instruction cycle. Timing and Control Memory-Reference Instructions, Input-Output and interrupt.

UNIT - IV	REGISTER TRANSFER AND MICRO-OPERATIONS	CLASSES: 11

Central processing unit: Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Complex Instruction Set Computer (CISC), Reduced Instruction Set Computer (RISC), Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit.
Pipeline and Parallel Processing: Parallel processing, Pipelining, Arithmetic pipeline, Instruction pipeline

UNIT - V	MEMORY SYSTEM	CLASSES: 11

INPUT-OUTPUT ORGANIZATION: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA. MEMORY ORGANIZATION: Memory Hierarchy, Main Memory, Auxiliary memory, Associate memory, Cache memory.
TEXT BOOKS
1. Digital Design- Morris Mano, PHI, 3rd Edition. 2. Digital Principles and Applications by Leach, Paul Malvino, 5th Edition. 3. Computer System Architecture- M. Morris Mano, 3rd edition, Pearson/PHI, India.
REFERENCE BOOKS
1. Switching and Finite Automata Theory by Zvi.Kohavi, Tata McGraw Hill. 2. Fundamentals of Digital circuits, A. Anand Kumar, Third Edition, 2013, PHI. 3. Computer Organization and Architecture William Stallings Sixth Edition, Pearson/PHI 4. Structured Computer Organization Andrew S. Tanenbaum, 4th Edition PHI/Pearson